

Machine Emulator application to run ROM games off the Raspberry Pi. The final appearance of the rig was like that of an old school retro gaming arcade and could play a variety of games off its local storage. The arcade machine even had a 32 button joystick between two players for a full on gaming experience. Interestingly enough, other makers have made miniature versions of the arcade Pi game machine that can fit in the palm of your hand.

Pi Supercomputer

The idea for a Pi supercomputer is fairly intuitive - what would happen if you combined a number of Pi boards together? Would it equal the power of a supercomputer? Well yes and no. So far the largest ever Pi based computer has been using 64 Raspberry Pi modules. This project was conducted at the University of Southampton, where a professor used Lego bricks to house and organise his Pi boards. These 64 boards were connected using their Ethernet ports and used a message passing interface to link up the processors over 32 nodes, allowing for efficient task allocation.



Lego bricks are a great way to set up a multi-Pi system.

The performance boost recorded was linear on the 700 MHz ARM processor per board and a total of 1TB of memory, with remarkable stability, limited only by the quality of the Ethernet network switches used. The official benchmark results of this project indicated a peak performance of 10.13 GFLOPS, which may not seem supercomputer grade today, but its worth remembering that the first supercomputer - the Cray-2 in 1985 clocked only 1.9 GFLOPS at the price of USD 16 million. The Pi supercomputer cost only about USD 3,500.

Gameboy Pi

Another gaming related project was the creation of a Nintendo Gameboy. The device used a discarded gameboy body to host the Pi and the software.